

2023 AMC 10B Problem 22 - Solution

Review the full statement and step-by-step solution for 2023 AMC 10B Problem 22. Great practice for AMC 10, AMC 12, AIME, and other math contests

2023 AMC 10B Problem 22

Problem:

How many distinct values of x satisfy $\lfloor x \rfloor^2 - 3x + 2 = 0$, where $\lfloor x \rfloor$ denotes the greatest integer less than or equal to x ?

Answer Choices:

- A. an infinite number
- B. 4
- C. 2
- D. 3
- E. 0

Solution:

First consider the possibility of integer solutions $x = k$. Then $k^2 - 3k + 2 = (k - 2)(k - 1) = 0$ has solutions $k = 1$ and $k = 2$.

Next consider the possibility of a noninteger solution x where $k < x < k + 1$ for some integer k . Then $\lfloor x \rfloor^2 - 3x + 2 = k^2 - 3x + 2 = 0$, which means $x = \frac{k^2 + 2}{3}$ with $k < x < k + 1$. Because $\frac{k^2 + 2}{3} > 0$, only nonnegative values of k need to be considered. The condition $k < \frac{k^2 + 2}{3}$ is equivalent to $0 < k^2 - 3k + 2$, which is true if and only if $k < 1$ or $k > 2$. The only nonnegative value with $k < 1$ is $k = 0$, leading to the solution $x = \frac{2}{3}$. The condition $\frac{k^2 + 2}{3} < k + 1$ is equivalent to $k^2 - 3k - 1 < 0$. The roots of $k^2 - 3k - 1$ are $k = \frac{3 \pm \sqrt{13}}{2}$, so $k^2 - 3k - 1 < 0$ if and only if

$$-0.3 \approx \frac{3 - \sqrt{13}}{2} < k < \frac{3 + \sqrt{13}}{2} \approx 3.3.$$

The only new integer value is $k = 3$, leading to the fourth solution $x = \frac{11}{3}$.

OR

Because $\lfloor x \rfloor^2$ is a nonnegative integer, $3x - 2$ must also be a nonnegative integer, so $x = \frac{n}{3}$ for some integer $n \geq 2$. Furthermore, if $\lfloor x \rfloor \geq 4$, then $\lfloor x \rfloor^2 - 3x + 2 = \lfloor x \rfloor^2 - 3(x - 1) - 1 \geq \lfloor x \rfloor^2 - 3\lfloor x \rfloor - 1 = \lfloor x \rfloor(\lfloor x \rfloor - 3) - 1 \geq \lfloor x \rfloor - 1 \geq 3 > 0$, so $2 \leq n \leq 11$ and $\lfloor x \rfloor^2 = 3x - 2 = n - 2$. The only perfect square values of $n - 2$ in this range are 0, 1, 4, and 9, occurring when $x = \frac{2}{3}, 1, 2$, and $\frac{11}{3}$, respectively. Each of these values of x satisfies the equation, so there are (B) 4 such values.

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